Quiz 5

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[redacted] Apr 30, 2021 [redacted]
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Instructions

You have exactly two hours to do this quiz. It is an open-book quiz, and you can use EP2 material given to you this semester (problem-solving solutions, lecture notes, etc). but you are not allowed to use the internet to solve the problems. You can use calculators for math and matrix calculations. If you need help with solving integrals in this quiz - you are allowed to use free integral solving sites, such as **https://www.integral-calculator.com/ (https://www.integral-calculator.com/)** . Needless to say, you also are not allowed to collaborate with anyone or seek help from anyone in solving these problems.

If you have technical problems: immediately contact the KSU help desk and let me know. You can reach me on zoom (**https://ksu.zoom.us/j/95088882288 (https://ksu.zoom.us/j/95088882288)**), discord, and email: **maravin@gmail.com (mailto:maravin@gmail.com)**

This quiz was locked May 16, 2021 at 11:59pm.

Attempt History

	Attempt	Time	Score
[redacted]			

[redacted]
Submitted Apr 30, 2021 at 6:30pm
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Question 1 **1 / 1 pts**

EP2 is almost over. Good luck!

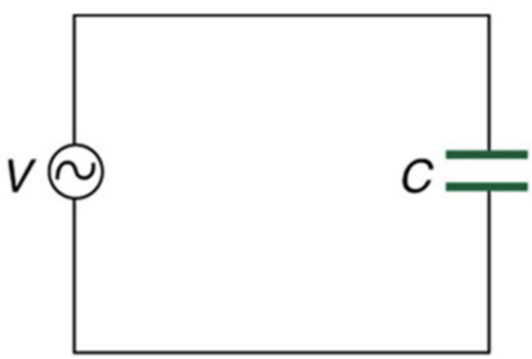
Ah, a question (sorry, not a multiple choice one): Are you ready to ace this quiz???

Correct!

Yes.

Question 2

3 / 3 pts



An AC voltage is applied across a capacitor. Initially, the current is flowing up. At that moment, what is the orientation of the magnetic field between the capacitor plates

- Up
- Down
- Clockwise looking from above

Correct!

- Counterclockwise looking from above
- There is no magnetic field

Question 3

3 / 3 pts

An electromagnetic wave is propagating into the screen (page). At a given point of time and space, the electric field points up. Where does the magnetic field point?

- up
- down
- left

Correct!

- right
- into the screen/page
- out of the screen/page

Question 4

3 / 3 pts

A vertically polarized light falls on a polarizer with a transmission plane oriented at 45 degrees with respect to vertical. If the original amplitude of the electric field was 1 V/m, what is the amplitude of the electric field of the wave after it emerges from the polarizer?

- 1 V/m
- Correct! 0.7 N/C
- 0.5 V/m
- 0.3 N/C
- 0.5 W/m^2

Question 5

3 / 3 pts

While digging your yard without checking for underlying cables you cut through the fiber-optics AT&T pipe and your backyard is now prettily illuminated with blue lights emanating from the severed cable. What is the frequency of the AT&T transmission signal?

- About 400 nm
- About 7.5×10^{14}
- Correct! About 7.5×10^{14} s^{-1}
- About 400 Hz
- About 7.5 m

Question 6

3 / 3 pts

While observing the ruined AT&T fiber optic cable from the previous question you wonder about the physical phenomenon that makes fiber optic signal transmission possible. What is it?

- dispersion
- polarization by reflection
- small angle refraction
- Correct! total internal reflection

Question 7

3 / 3 pts

What kind of image does a plane mirror create?

- real, upright
- Correct! virtual, upright
- real, inverted
- virtual, inverted
- either real or virtual, depending on the distance, always upright.

Question 8

3 / 3 pts

You have a concave mirror with a radius of curvature of 10 cm. Where do you need to place the object to produce a real image of the object that is neither magnified nor demagnified?

Correct!

- p = 10 cm
- p = 5 cm
- p = 0 cm
- p = -5 cm
- p = -10 cm

Question 9

3 / 3 pts

A mystery mirror can produce real and virtual images. What kind of mirror is that?

- convex

Correct!

- concave
- plain
- plain or concave
- plain or convex

Question 10

3 / 3 pts

Another mystery mirror can produce only virtual images. What kind of mirror is that?

- convex
- concave
- plain
- plane or concave
- Correct! plane or convex

Question 11

3 / 3 pts

A convex mirror of a radius of 20 cm produces an image of an object placed at an unknown distance **p** from the mirror. The object itself is 2 cm in height, while the image is 1 cm. What is the value of **p**?

- 20 cm
- Correct! 10 cm
- 5 cm
- -5 cm
- -10 cm
- None of the above

Question 12

3 / 3 pts

An old person is wearing reading glasses while reading a book (kind of an iPad but without batteries and need for wi-fi). While reading a book, glasses produce

Correct!

- virtual images
- real images
- no images

Question 13

0 / 3 pts

A lens is made from a material with n = 1.5 and it makes an image of the sun at the distance of 10 cm from the lens. If you place the lens in the isopropyl alcohol that has n = 1.5, where will the image of the sun be?

- Same distance, 10 cm

ou Answered

- Somewhat closer to the lens, smaller than 10 cm
- A bit further away

orrect Answer

- Nowhere, the lens will not converge light
- What kind of questions are these???

Question 14

3 / 3 pts

Which lens will **always** produce an image that is closer to the lens than the object?

- it is not possible
- converging
- Correct! diverging
- can be either converging or diverging

Question 15

3 / 3 pts

You are looking at a tiny insect, trying to determine whether it is dangerous or not. To see it better, you should

- bring it as close as possible to your eye without touching it
- Correct! get a lens with $f > 0$ and look at the insect through it
- get a lens with $f < 0$ and look at the insect through it
- ask your favorite prof with a microscope to do it and send you a report instead

Question 16

3 / 3 pts

You have a laser beam with λ = 400 nm that you split into two independent secondary beams of light. One of the secondary light beams travels about 4 cm further than the other one before they reach the same spot on the screen. The intensity of the light on the screen will be

- zero
- the same as the intensity of the original, primary beam
- Correct! twice the intensity of the original, primary beam
- four times the intensity of the original, primary beam

Question 17

3 / 3 pts

Two laser lights with λ = 650 nm that were split from the original laser beam travel to the same spot on the screen. One of the beams travels 4.875 µm further than the other. The intensity of the light on the screen will be

- Correct! zero
- the same as that of either beam by itself
- double that of either beam by itself
- Four times that of either beam by itself

Question 18

3 / 3 pts

Suppose you replaced the two lights in the Young experiment with lights of the same wavelength, but one of the lights is unpolarized, while the other one is vertically polarized. Will you be able to see the interference pattern?

Correct!

- No
- Yes
- Not enough information to answer this question

Question 19

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Evil Professor Maravin, whose height is 1.9 m, stands 2 m in front of a concave spherical mirror that has a radius of curvature of 5 m. Find

A. (3 points) Focal length of the mirror, including the sign.
B. (2 points) Location of Evil Professor Maravin's image.
C. (2 points) Virtuality and inversion state of Evil Professor Maravin's image
D. (2 points) Height of Maravin's image
E. (5 points) Check your answer with a ray diagram. *One bonus point if you use all four valid rays.*

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Question 20

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A thin symmetric lens has a focal length f = -0.2 m. An object of height 10 cm is placed at a distance of 40 cm to the left of the lens. Determine the following:

A. (4 points) The distance of the image to the lens and whether the image is on the left or right of the lens.
B. (3 points) The height of the image, its virtuality, and the inversion state.
C. (5 points) Check your answers with a ray diagram. *One bonus point if you use three valid rays.*

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Question 21

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Dr. Maravin's eyes have lenses that can change their focal distance between 1.693 cm and 1.631 cm. The effective distance between his eye's lenses and the retina where the image is formed is 1.7 cm.

A. (3 points) Find the closest distance where Dr. Maravin's eyes can focus on (near point).
B. (3 points) Find the furthest distance where Dr. Maravin's eyes can focus on (far point).
C. (4 points) Find the focal distance of the corrective contact lens that would allow Dr. Maravin to see distant starts sharply.
D. (4 points) Find the focal distance of the corrective contact lens that would allow Dr. Maravin to focus on this screen from a distance of 0.25 m.

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Question 22

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In one of the obscure experiments, a deep-diver was asked to dive into a deep pool filled with oil (n = 1.46). When the deep-diver reached the bottom of the oil tank and started his ascent toward the surface, he noticed that he could only see the world outside the oil tank in a circular spot right above his head - the surface of the oil was dark outside the light spot.

A. (6 points) Determine the radius *R* of the circular spot when the deep-diver is at the depth of 10 meters.
B. (2 points) As the deep-diver raises to the surface, how does *R* change? (Stays the same, increase, decrease?)

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